

JKBOSE 2026

CLASS 11TH

STATISTICS

MATHS

LECTURE-3

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TOPICS TO BE COVERED

1

Variance and Standard Deviation

2

Related Problems

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N}, \quad \bar{x} = \frac{\sum x_i}{n}$$

$$\overline{x^2} = \frac{\sum_{i=1}^n f_i x_i^2}{N}$$

$$M = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$\frac{\text{Upper limit} + \text{lower limit}}{2}$$

Median class $\rightarrow$ cf is $\frac{N}{2}$ or Just greater than $\frac{N}{2}$.

EXAMPLE

Calculate the **mean deviation** about **median** for the following data:

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	6	7	15	16	4	2

(Sol) Here $N = 50 \Rightarrow \frac{N}{2} = 25$. So 20-30 is the Median class. Then the median is given by;

$$M = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h = 20 + \left(\frac{25 - 13}{15} \right) \times 10$$

$$\Rightarrow M = 20 + \frac{12}{15} \times 10 = 20 + 8 = 28$$

So, Median, $M = 28$.

$$\text{Therefore } M.D(M) = \frac{\sum_{i=1}^n f_i |x_i - M|}{N} = \frac{508}{50}$$

$$\Rightarrow M.D(M) = 10.16$$

Hence the result.

Class	frequency(f_i)	Cf	Class mark(x_i)	$x_i - M$	$ x_i - M $	$f_i x_i - M $
0-10	6	6	5	-23	23	138
10-20	7	13	15	-13	13	91
20-30	15	28	25	-3	3	45
30-40	16	44	35	7	7	112
40-50	4	48	45	17	17	68
50-60	2	50	55	27	27	54
	50					508

Median Class →

$$M = 28$$

EXERCISE – 13.1

QUESTION

Find the mean deviation about median for the follow data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of Girls	6	8	14	16	4	2

H/W

QUESTION

Calculate the mean deviation about median age for the age distribution of 100 persons given below: $\frac{1}{2} = 0.5$

Age (in years)	16-20	21-25	26-30	31-35	36-40	41-45	46-50	51-55
Number	5	6	12	14	26	12	16	9

HINT: [Convert the given data into continuous frequency distribution by subtracting 0.5 from the lower limit and adding 0.5 to the upper limit of each class interval]

Age (in years)	15.5-20.5	20.5-25.5	25.5-30.5	30.5-35.5	35.5-40.5	40.5-45.5	45.5-50.5	50.5-55.5
Number	5	6	12	14	26	12	16	9

Variance and Standard Deviation



Variance: The Mean of the Squares of the deviations from Mean is known as a variance. It's generally denoted by σ^2 .

For ungrouped data;

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

For Grouped data;
$$\sigma^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{N}$$

* The positive square root of the variance is known as the standard deviation (σ).

$$\text{Here } \sigma^2 = \frac{\sum_{i=1}^n f_i (x_i^2 + \bar{x}^2 - 2\bar{x}x_i)}{N}$$

$$\Rightarrow \sigma^2 = \frac{1}{N} \left\{ \sum_{i=1}^n f_i x_i^2 + \sum_{i=1}^n f_i \bar{x}^2 - \sum_{i=1}^n 2\bar{x} f_i x_i \right\}$$

$$\Rightarrow \sigma^2 = \frac{1}{N} \left\{ \sum_{i=1}^n f_i x_i^2 + \bar{x}^2 \sum_{i=1}^n f_i - 2\bar{x} \sum_{i=1}^n f_i x_i \right\}$$

$$\Rightarrow \sigma^2 = \frac{1}{N} \left\{ \sum_{i=1}^n f_i x_i^2 + \bar{x}^2 N - 2\bar{x} \sum_{i=1}^n f_i x_i \right\}$$

$$\Rightarrow \sigma^2 = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 + \cancel{\frac{1}{N} \bar{x}^2 N} - 2\bar{x} \left(\frac{1}{N} \sum_{i=1}^n f_i x_i \right)$$

$$\Rightarrow \sigma^2 = \overline{(x^2)} + (\bar{x})^2 - 2\bar{x} \times \bar{x}$$

$$\Rightarrow \sigma^2 = \overline{(x^2)} + (\bar{x})^2 - 2(\bar{x})^2$$

$$\Rightarrow \sigma^2 = \overline{(x^2)} - (\bar{x})^2$$

$$\text{i.e. } \sigma^2 = (\text{Mean of } x^2) - (\text{Mean of } x)^2$$

* Variance and Standard Deviation for first n — natural numbers:

Observation: 1, 2, 3, 4, 5, 6, ..., n .

Then their variance is given by;

$$\Rightarrow \sigma^2 = \left(\frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n} \right) - \left(\frac{1 + 2 + 3 + \dots + n}{n} \right)^2$$

$$\Rightarrow \sigma^2 = \left(\frac{\frac{n(n+1)(2n+1)}{6}}{n} \right) - \left(\frac{\frac{n(n+1)}{2}}{n} \right)^2$$

$$\Rightarrow \sigma^2 = \frac{\cancel{n}(n+1)(2n+1)}{\cancel{6}\cancel{n}} - \left(\frac{\cancel{n}(n+1)}{\cancel{2}\cancel{n}} \right)^2$$

$$\Rightarrow \sigma^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)}{2} \left\{ \frac{(2n+1)}{3} - \frac{(n+1)}{2} \right\}$$

$$\Rightarrow \sigma^2 = \left(\frac{n+1}{2} \right) \left(\frac{4n+2-3n-3}{6} \right)$$

$$\Rightarrow \sigma^2 = \frac{(n+1)(n-1)}{12}$$

$$\Rightarrow \sigma^2 = \frac{(n^2-1)}{12}$$

Also, the $\text{S.D}(\sigma) = \sqrt{\frac{n^2-1}{12}}$

QUESTION (NCERT EXAMPLER)

The standard deviation of the data 6, 5, 9, 13, 12, 8, 10 is.

1 $\sqrt{\frac{52}{7}}$

2 $\frac{52}{7}$

3 $\sqrt{6}$

4 6

(Sol) Here $\bar{x} = \frac{6+5+9+13+12+8+10}{7}$

$\Rightarrow \bar{x} = \frac{63}{7} \Rightarrow \bar{x} = 9$

Therefore $\sigma^2 = \frac{(6-9)^2 + (5-9)^2 + (9-9)^2 + (13-9)^2 + (12-9)^2 + (8-9)^2 + (10-9)^2}{7}$

$\Rightarrow \sigma^2 = \frac{9+16+0+9+9+1+1}{7}$

$\Rightarrow \sigma^2 = \frac{32}{7}$

$\Rightarrow \sigma = \sqrt{\frac{32}{7}}$ (Standard deviation)

QUESTION (NCERT EXAMPLER)

Standard deviations for first 10 natural numbers is.

1 5.5 ✗

2 3.87 ✗

3 2.97 ✗

4 2.87 ✓

$$\sigma = \sqrt{\frac{10^2 - 1}{12}}$$

$$\Rightarrow \sigma = \sqrt{\frac{99}{12}}$$

$$\sigma^2 = \frac{10^2 - 1}{12}$$

$$\sigma = \sqrt{\frac{10^2 - 1}{12}}$$

QUESTION (NCERT EXAMPLER)

If the variance of a data is 121, then the standard deviation of the data is 11.

$$\sigma^2 = 121$$

$$\sigma = \sqrt{121} = 11$$

$$\frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \frac{\sum f_i x_i^2}{N} + \frac{(\bar{x})^2 \sum f_i}{N} - 2 \bar{x} \frac{\sum f_i x_i}{N} \quad \text{H/W}$$

EXAMPLE

Find the variance and standard deviation for the following data:

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

(Sol) Here $\bar{x} = \frac{420}{30} = 14 \Rightarrow \bar{x} = 14$.

Also, $\sigma^2 = \frac{1374}{30} = 45.8$.

& $\sigma = \sqrt{45.8} = 6.77$

Therefore variance = 45.8

& Standard deviation = 6.77.

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i (x_i - \bar{x})^2$
4	3	12	-10	100	300
8	5	40	-6	36	180
11	9	99	-3	9	81
17	5	85	3	9	45
20	4	80	6	36	144
24	3	72	10	100	300
32	1	32	18	324	324
	30	420			1374

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Thank You